ESP32 Wi-Fi Channel State Information Acquisition and Exploratory Signal Visualization for Human Presence Sensing

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Abstract. Wi-Fi Channel State Information (CSI) provides a potential basis for device-free sensing because changes in wireless propagation can be observed through channel measurements. However, CSI is affected not only by people but also by device placement, furniture, metallic objects, walls, and other changes in the propagation environment. Consequently, signal variation alone does not establish human presence, and performance in an unfamiliar environment must be evaluated experimentally.

This paper presents an ESP32-based CSI acquisition and exploratory visualization prototype. The supplied log was parsed as comma-separated records, with the quoted CSI array retained as a single field. The log contains 16,660 records, each with 25 parsed fields. The examined channel-related fields contain the value 11 throughout the log. Received signal strength indicator (RSSI) values range from -73 to -29 dBm, with a mean of approximately -42.45 dBm. Each record reports a CSI-length value of 256 and contains an array of 256 integer values. The exact interpretation of this array requires confirmation against the ESP32 model and firmware configuration.

The available log contains no verified human-presence labels or environment identifiers. In addition, the supplied dashboard screenshot indicates that calibration is required and that receiver data are unavailable in the displayed state. Accordingly, this work reports descriptive properties of the collected log and documents the visualization interface; it does

not claim a validated human detector, detection accuracy, or robustness to environmental changes. A reproducible protocol for future labelled, environment-wise evaluation is provided.

Keywords: Wi-Fi sensing · Channel state information · ESP32 · Human presence sensing · Signal visualization · Environment-wise evaluation

1 Introduction

Human presence sensing has potential applications in occupancy monitoring, smart buildings, and assistive systems. Conventional sensors can require dedicated hardware or direct visibility of the target. Wi-Fi sensing is an alternative research direction in which changes in wireless propagation are examined to infer activity or environmental changes.

Channel State Information (CSI) describes a wireless channel estimate associated with the measured transmission paths and subcarriers. Changes in the propagation environment can alter the measured CSI. Human movement may contribute to these changes, but so can moving objects, changes in device position, interference, and multipath variation. CSI changes are therefore not uniquely attributable to a person [1, 2].

Low-cost ESP32 hardware provides an accessible platform for investigating CSI acquisition. Nevertheless, a functioning acquisition pipeline and a signal visualization interface are not, by themselves, evidence of reliable human-presence classification. A detector must be evaluated against independently recorded labels and under clearly specified test conditions.

This paper describes an ESP32 CSI logging prototype, summarizes the observed properties of the supplied log, and documents the scope and limitations of the accompanying dashboard. It also outlines an evaluation protocol for subsequent human-presence experiments.

The contributions of this paper are:

- a description of an ESP32-based CSI logging and visualization workflow;
- a descriptive summary of the supplied CSI log, including record count, parsed field count, channel-related field values, and RSSI statistics;
- an explicit separation between observed data, proposed signal-processing methods, and experiments that remain to be conducted; and
- a reproducible plan for evaluating human-presence sensing using labelled recordings and held-out environments.

The present work is a prototype and exploratory data study. It does not claim that a human-presence classifier has been trained or validated.

2 Background and Related Work

Ma et al. survey Wi-Fi sensing methods that use CSI for sensing and recognition tasks, describing signal-processing approaches and practical challenges [1]. Miao

et al. review Wi-Fi sensing techniques for human activity recognition and discuss challenges associated with practical sensing conditions [2]. Quy et al. investigate a method that combines CSI amplitude and phase information with attention-based feature fusion for activity recognition [3].

These studies motivate examining multiple signal representations and evaluating sensing methods beyond a single recording condition. However, activity recognition and binary human-presence detection are different tasks. Results from an activity-recognition method cannot be treated as evidence that the present prototype detects human presence.

The present study focuses on acquisition and visualization as preliminary stages. Reference-based processing and environment-wise classification are discussed as future experimental directions, rather than as completed or validated components.

3 Prototype and Data Acquisition

3.1 System Overview

The intended sensing arrangement uses Wi-Fi transmissions and an ESP32 device configured to acquire CSI at the receiving side. A logging process stores the received records for subsequent inspection, while a host-side dashboard presents signal summaries and visualizations.

The precise ESP32 model, firmware version, transmitter configuration, receiver configuration, packet-generation method, and CSI decoding convention should be recorded with the experimental implementation. These details are necessary to reproduce the measurements and to interpret the integer array correctly. They are not inferred here where the supplied materials do not establish them.

The intended processing stages are illustrated in Fig. 1. The classification and validation stages represent a proposed workflow; their presence in the diagram does not imply that they have already been implemented or evaluated.

3.2 Dashboard Interface

The supplied dashboard screenshot is shown in Fig. 2. The interface presents signal-oriented summaries and visualizations, including RSSI, amplitude-related summaries, a motion indicator, packet correlation, a subcarrier heatmap, and a fluctuation spectrum.

These names describe the displayed interface. The available evidence does not independently establish the precise formula, normalization, window length, or calibration procedure used for every dashboard indicator. Such implementation details should be documented before the indicators are interpreted as validated features.

The screenshot reports that calibration is required and that receiver data are unavailable in the displayed state. It is therefore treated as documentation of the interface, not as proof of continuous live acquisition or a successful detection event.

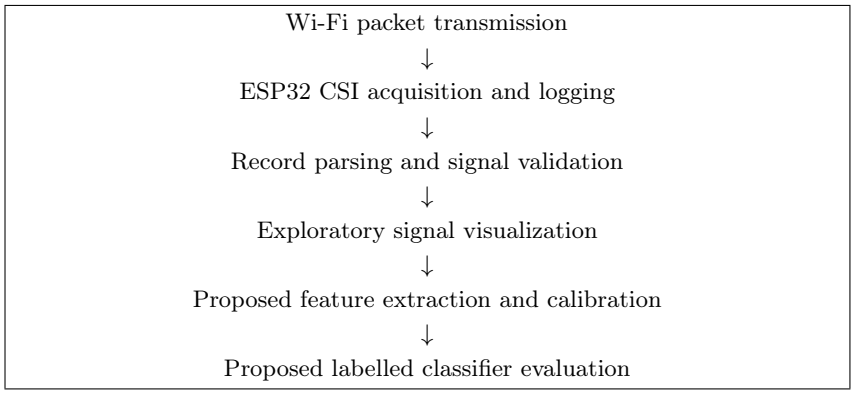


Fig. 1. Prototype workflow and proposed subsequent evaluation stages. The final classification stage requires labelled experiments and is not reported as validated in this study.

3.3 Hardware Documentation

If the following images are present in the Overleaf project, they can document the physical prototype. Their captions should be updated if the actual hardware arrangement differs from the descriptions below.

4 Log Parsing and Descriptive Results

4.1 Parsing Method

The supplied `run1.log` file contains one CSI record per line. Each line includes a comma-separated prefix and a quoted array containing the CSI-related integer values. The array contains commas internally, so splitting a line naively at every comma produces an incorrect field count. Parsing the records with a CSV-aware parser retains the quoted array as one field.

Under this parsing method, all 16,660 records contain 25 fields. The record marker is `CSI_DATA`. The channel-related fields examined in the log contain the value 11 in every record. The RSSI values and CSI array lengths were calculated from the parsed fields.

The values in Table 1 describe the supplied file. The RSSI mean is a descriptive statistic over the records in that file; it is not a detection threshold or evidence of a human-presence event. Similarly, the channel-related field value does not establish that the experiment was repeated across different channels or environments.

4.2 Interpretation of the CSI Array

The log reports a CSI-length value of 256, and the quoted array in each record contains 256 integers. This establishes the stored length, but not the physical

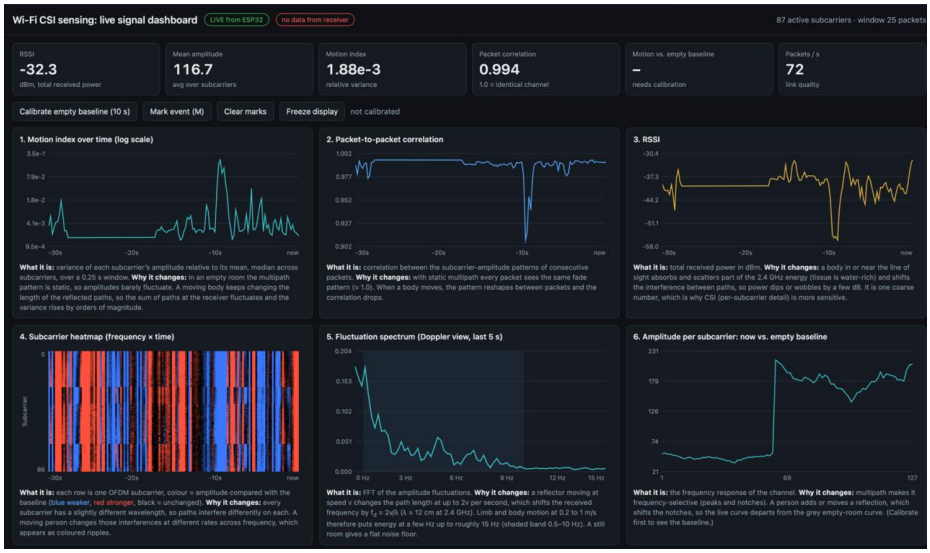


Fig. 2. Supplied prototype dashboard. The screenshot documents the interface and its displayed status; it is not a quantitative evaluation of human-presence detection.

meaning of every element. Depending on the device and firmware representation, CSI data may encode components that require device-specific interpretation before amplitude, phase, or subcarrier-level features can be calculated.

For that reason, the array is not treated in this paper as a confirmed sequence of 256 independent subcarriers. The dashboard's displayed count of active subcarriers must also not be equated with the logged array length without documenting the conversion and indexing rules used by the implementation.

Before deriving physical features from the raw array, the implementation should specify:

- the ESP32 model and the CSI format documented for that model;
- the interpretation and ordering of the array elements;
- whether the array contains interleaved in-phase and quadrature components or another representation;
- any invalid, unused, or excluded entries; and
- the mapping between raw entries and the subcarriers displayed by the dashboard.

These checks are necessary for reproducible signal processing and should not be replaced by assumptions based solely on the array length.

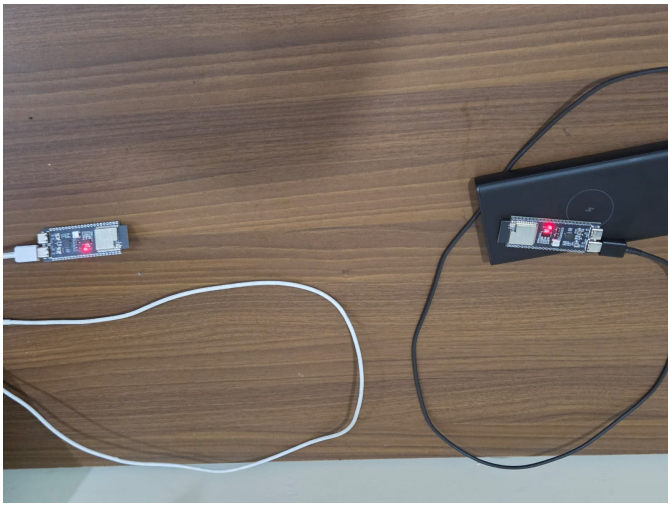


Fig. 3. ESP32 hardware used in the prototype, if the corresponding image is supplied. The exact board model should be identified in the implementation description.

5 Signal Representation and Proposed Processing

This section defines candidate operations for subsequent analysis. They describe a possible processing method; they are not presented as calculations already applied to the supplied log.

5.1 Complex CSI, Amplitude, and Phase

If the device-specific format has been verified to provide compatible in-phase and quadrature components, the complex channel estimate for subcarrier k at sample index t can be represented as

$$H_k(t) = I_k(t) + jQ_k(t), \tag{1}$$

where $I_k(t)$ and $Q_k(t)$ denote the in-phase and quadrature components, respectively.

The amplitude and phase are then

$$A_k(t) = \sqrt{I_k(t)^2 + Q_k(t)^2}, \tag{2}$$

and

$$\phi_k(t) = \operatorname{atan2}\big(Q_k(t), I_k(t)\big). \tag{3}$$

Equations 1–3 are applicable only after the raw data representation has been verified. The present descriptive statistics do not establish that these equations were applied to the supplied array.



Fig. 4. Prototype setup, if the corresponding image is supplied. The transmitter and receiver roles, device separation, and environment should be stated in the final manuscript only after they have been verified.

Table 1. Descriptive statistics of the supplied log.

Property	**Observed value**
Number of records	16,660
Parsed fields per record	25
Record marker	`CSI_DATA`
Channel-related field values	11 throughout the log
Minimum RSSI	−73 dBm
Maximum RSSI	−29 dBm
Mean RSSI	Approximately −42.45 dBm
Reported CSI-length field	256 in every record
Parsed CSI-array length	256 integers in every record
Human-presence labels	Not available
Environment or room labels	Not available

5.2 Reference-Based Features

A future experiment may compare features extracted directly from validated CSI measurements with features computed relative to an empty-environment reference. Let $A_{e,k}(t)$ denote the validated amplitude for subcarrier k in environment e. For N reference samples, define the reference mean as

$$B_{e,k} = \frac{1}{N} \sum_{t=1}^{N} A_{e,k}(t). \tag{4}$$

A reference-relative feature can then be defined as

$$D_{e,k}(t) = A_{e,k}(t) - B_{e,k}. \tag{5}$$

This subtraction measures deviations from a reference. It does not guarantee the removal of environmental effects, and it cannot, by itself, uniquely distinguish

a person from every other change in the propagation environment. The reference must be collected under a documented empty-room condition if it is intended to represent that condition.

5.3 Candidate Temporal Features

For a window containing T samples, the mean amplitude for subcarrier k can be defined as

$$\mu_k = \frac{1}{T} \sum_{t=1}^{T} A_k(t), \tag{6}$$

and the corresponding population variance as

$$\sigma_k^2 = \frac{1}{T} \sum_{t=1}^{T} \left(A_k(t) - \mu_k \right)^2 . \tag{7}$$

Other candidate features include temporal differences, standard deviation, and cross-subcarrier correlation. These features require a validated CSI representation, an explicitly defined sampling window, and a documented preprocessing procedure.

The supplied dashboard includes motion and correlation indicators, but their exact implementation formulas have not been established from the log alone. They are therefore treated as exploratory visualization indicators rather than validated human-presence features.

6 Evaluation Protocol for Future Experiments

6.1 Ground-Truth Collection

A human-presence detector requires labels that are recorded independently of the CSI values. Each experimental recording should be associated with a documented label and metadata describing the test condition.

The proposed dataset should include:

- empty-room recordings;
- recordings with a person present but stationary;
- recordings with controlled human movement;
- empty-room recordings containing stationary metallic objects;
- empty-room recordings involving object, door, or furniture movement; and
- recordings from at least one environment held out from training and model selection.

Each session should record the environment identifier, device positions, approximate separation, wall or obstruction conditions where relevant, channel configuration, packet-generation method, recording duration, and label-annotation procedure. These values must be measured or documented during collection rather than reconstructed from unlabelled CSI records.

If through-wall sensing is to be claimed, line-of-sight and through-wall tests must be evaluated separately, with the relevant wall conditions documented.

6.2 Environment-Wise Data Splitting

Randomly assigning individual packets from one recording to both training and test sets can result in highly correlated samples appearing in both sets. Such a split may overstate performance on unseen conditions.

The proposed evaluation should split data by recording session and, for the principal generalization experiment, by environment. Model selection should use only training and validation environments. The held-out environment should be reserved for final evaluation.

If deployment requires empty-room calibration, calibration-based and calibration-free configurations should be reported separately. The duration and sample count of the reference recording should be specified. Test labels must not be used to tune preprocessing or model parameters.

6.3 Evaluation Metrics

After labelled data and a trained classifier become available, let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives. Precision and recall are defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{9}$$

The F1-score and false-positive rate are

$$\text{F1} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{10}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \tag{11}$$

These metrics are definitions for future evaluation; they cannot be calculated from the current unlabelled log. Implementations should also handle zero denominators explicitly.

Results should include the number of recordings, class balance, the unit of evaluation, and the split procedure. False positives should be examined separately for empty rooms containing stationary metallic objects and for conditions involving object movement.

7 Results and Discussion

The supplied file supports a descriptive account of CSI logging. It contains 16,660 parsed records, with 25 fields per record, a reported CSI length of 256, and a 256-integer array in every record. The examined channel-related fields contain the value 11 throughout the file. RSSI ranges from -73 to -29 dBm and has a mean of approximately -42.45 dBm.

These observations characterize the contents of the supplied log. They do not establish which records correspond to a person being present, whether the room was empty, or whether metallic objects or other environmental changes were introduced. No environment identifiers or independently recorded human-presence labels are available in the supplied file.

The dashboard screenshot demonstrates the existence of a visualization interface, but the displayed status indicates that calibration is required and receiver data are unavailable in that state. It should not be interpreted as a live detection result. Likewise, visible peaks or fluctuations in a signal plot cannot be attributed to human movement without time-aligned experimental annotations.

Consequently, the present study does not report a trained classifier, accuracy, precision, recall, F1-score, false-positive rate, or measured improvement from reference-based processing. No conclusion about generalization across rooms or robustness to metallic objects can be drawn from the available data.

A subsequent controlled study should compare conventional CSI features and reference-relative features on identical held-out recordings. The comparison should report both detection performance and empty-room false alarms. Negative or inconclusive results should be reported alongside positive findings.

8 Limitations

The available log is not labelled by human presence, room, or experimental condition. The dashboard screenshot reports that calibration is required and receiver data are unavailable in the displayed state. The available evidence therefore supports a description of a logging artifact and visualization interface, not a validated human-presence detector.

The exact mapping between the 256 integers in each CSI array and the underlying CSI representation must be verified against the specific ESP32 hardware and firmware. The dashboard's displayed active-subcarrier count should not be equated with the logged array length until this mapping is documented.

The supplied materials also do not establish a reproducible sampling rate, a fully specified feature-extraction implementation, a ground-truth annotation procedure, or performance on held-out environments. These details must be reported before classification results can be independently reproduced.

Finally, CSI changes can arise from multiple sources. Reference subtraction may reduce some persistent offsets but cannot, by itself, prove that a change was caused by a person. The paper does not claim validated localization, activity recognition, through-wall performance, or multi-person detection.

9 Conclusion

This paper describes an ESP32-based Wi-Fi CSI acquisition and exploratory visualization prototype and summarizes the supplied log using CSV-aware parsing. The log contains 16,660 records, each with 25 parsed fields, and each record

contains a 256-integer CSI-related array. The examined channel-related fields consistently contain 11, while the RSSI values range from -73 to -29 dBm, with a mean of approximately -42.45 dBm.

These findings are descriptive. Because the supplied data do not contain verified human-presence labels or environment identifiers, they do not establish the performance or environmental robustness of a detector. The dashboard screenshot documents the interface but does not provide evidence of validated live detection.

The next stage is to verify the raw CSI representation, collect labelled recordings across controlled conditions, and evaluate candidate features using held-out environments. Only those experiments can determine whether reference-based processing improves human-presence sensing under environmental variation.

References

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